

# TensPilot+

Design of a Dual-Mode TENS Device

*with a Companion Web Application*

# Presentation Outline

## Hardware

- The Problem
- Core Device Design
- Two Therapy Modes
- Testing & Validation

## Software

- Patient Application
- Cloud Synchronization
- Doctor Dashboard
- Gemini AI Assistant

## Impact

- Market Comparison
- Addressing SDGs
- Accessibility
- Conclusion

# 1. The Problem & Our Solution

## The Problem

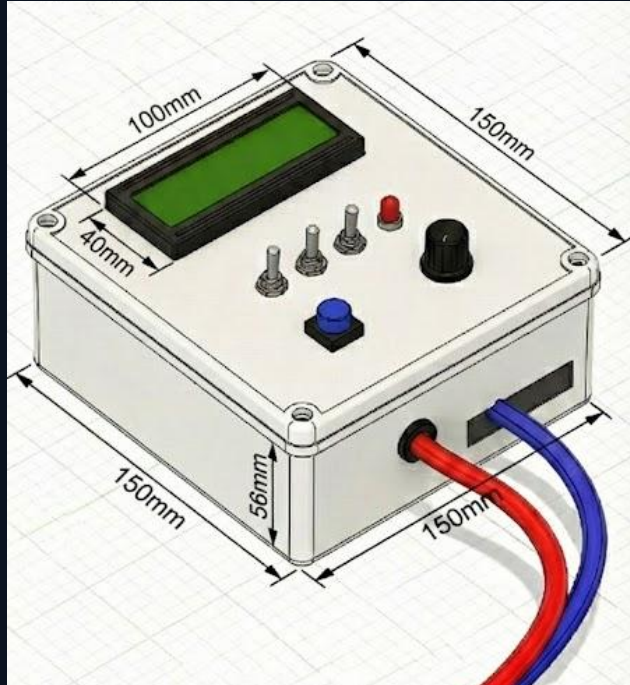
- Standard TENS devices provide effective pain relief but operate as standalone systems.
- No structured tracking of treatment parameters such as pulse width, frequency, or intensity.
- Patients often rely on memory or manual records to report therapy sessions.
- Clinicians have limited visibility into home-based treatment progress and outcomes.

## Our Solution

TensPilot+ combines a locally developed TENS device with a smart monitoring platform:

- Tracks treatment sessions and pain outcomes.
- Supports remote clinician oversight.
- Provides data-driven insights for better care.
- Makes home-based electrotherapy more effective and accountable.

## 2. Hardware Design & Testing



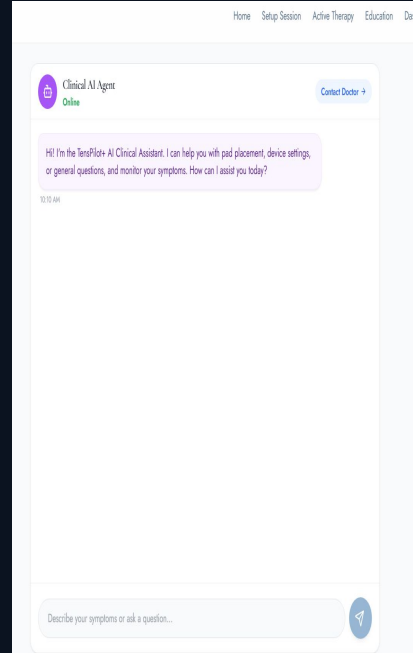
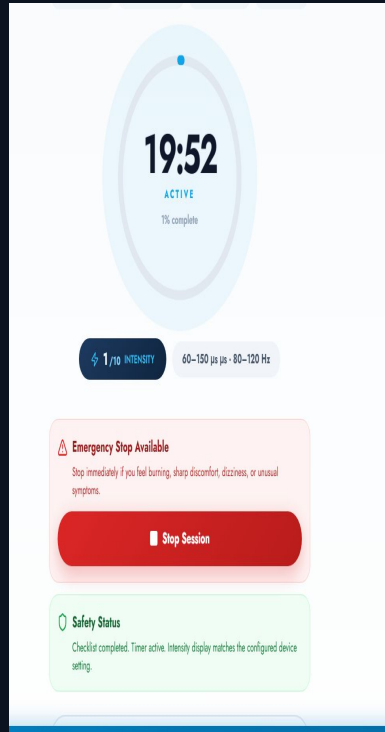
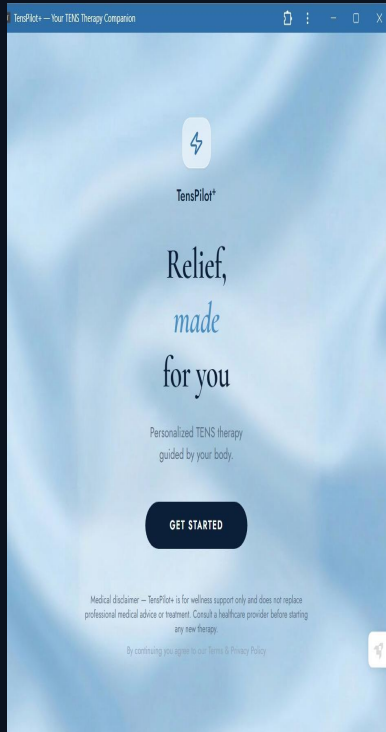
### Internal Components

- Microcontroller Unit (Arduino Uno)
- Battery Supply (5V Power Bank)
- L298N H-Bridge Driver
- Pulse Generation Circuit
- Hardware Safety Features

### Operational Capabilities

- Tested on 12 Volunteers
- Frequency (1–120 Hz) & Pulse Width Control
- Intensity (mA) Control
- Portable Operation (4+ Hours)

# 3. Ecosystem: Patient Application



## How the App Supports the Device

- Guides users through therapy sessions
- Records treatment parameters and outcomes
- Tracks pain improvement over time
- Connects patients with healthcare providers
- Enables access to AI-assisted support

The TensPilot+ Patient Application serves as the digital companion to the TENS device, helping users manage therapy sessions, monitor pain levels, receive guidance, and securely share treatment data with healthcare professionals.

# 4. Remote Monitoring & Clinical Oversight

The screenshot displays the 'Advanced TENS' section of the TensPilot+ interface. It features three main interactive areas: a 'Cloud Synchronization' section with a 'Sync to Cloud Now' button; a 'Link to Doctor' section with a text input for a 6-digit access code and a 'Link' button; and an 'Add Patient Profile' section with multiple input fields for patient details.

Profile      Advanced TENS

**Cloud Synchronization**  
Your profiles and TENS sessions are securely linked to your account (chukwumablessing031@gmail.com).  
[Sync to Cloud Now](#)

**Link to Doctor**  
Enter the 6-digit access code provided by your clinic to enable live therapy monitoring and video consultations.  
ENTER 6-DIGIT CODE... [Link](#)

**Add Patient Profile**

Patient name \*

Primary condition (e.g. Lower Back Pain)

Age

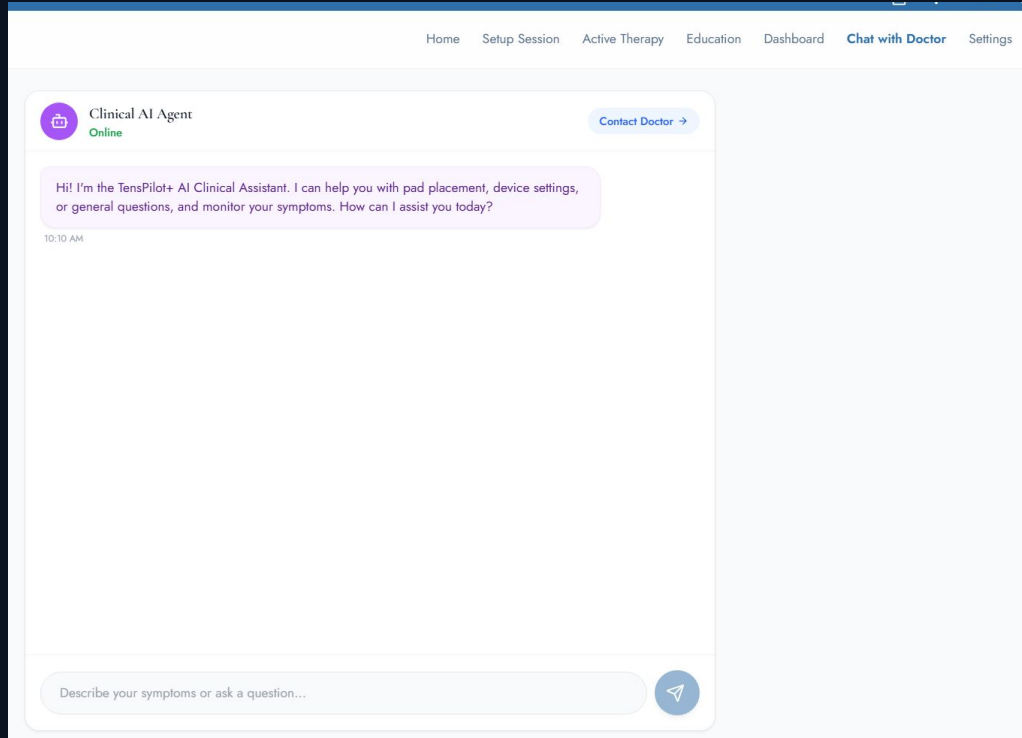
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Supervising Physician

## Clinical Benefits

- Secure storage of treatment records
- Connects patients directly with healthcare providers
- Monitors therapy usage and adherence
- Tracks pain improvement over time
- Supports informed treatment decisions

# 5. AI Integration: Gemini Virtual Assistant



## Gemini-Powered Capabilities

- Recommends electrode placement based on pain type.
- Provides therapy guidance 24/7.

## Key Benefits

- Reduces patient misinformation.
- Improves therapy adherence and confidence.

## 6. Two Therapy Modes

### CONVENTIONAL MODE

**20–120 Hz**

Fast pain relief (10–20 minutes)

For acute pain & muscle soreness

Smooth tingling sensation

### ACUPUNCTURE-LIKE MODE

**1–10 Hz**

Longer relief (90–180 minutes)

For chronic & neuropathic pain

Rhythmic tapping with contractions



## 7. Innovative Edge & Market Comparison

| Feature         | Imported Smart TENS      | TensPilot+             |
|-----------------|--------------------------|------------------------|
| Cost            | Highly Expensive         | Low Cost & Accessible  |
| Monitoring      | Limited / Bluetooth Only | Global Cloud Telemetry |
| AI Guidance     | None                     | Gemini Integrated      |
| Nigeria-Focused | No                       | Yes (Offline/PWA)      |

# 8. Addressing Sustainable Development Goals

## SDG 3: Good Health

Improved pain management & better quality of life.

## SDG 9: Industry & Innovation

AI-enabled healthcare innovation through local hardware.

## SDG 10: Reduced Inequalities

Affordable care for underserved populations & rural areas.

## SDG 17: Partnerships

Connects patients and clinicians digitally across regions.

## 9. Impact & Accessibility

### Designed for Nigeria

- ✓ Works Offline via Service Workers
- ✓ Runs on low-end Android phones
- ✓ Reduces transportation costs
- ✓ Suitable for rural healthcare

### Vulnerable Demographics

- Women: Chronic pelvic & menstrual pain.
- Adolescents and children under professional supervision
- Elderly: Arthritis care with reduced clinic visits.

# Thank You For Listening!

TensPilot+

*Any Questions?*